

Part 6

Build, test and ship

A short iteration loop, a test drive that catches problems in the right order, ways to inspect what the game really loaded, and a symptom-to-fix table for the whole pipeline.

In this part

1. The iteration loop: what to redo after each kind of change
2. Getting the vehicle into the game
3. The test drive, in order
4. Looking inside compiled files
5. Shipping
6. Troubleshooting: symptom → cause → fix

Steps inside the game's editors vary slightly between TSC releases; where exact menu names matter they are marked

check.

1 · The iteration loop

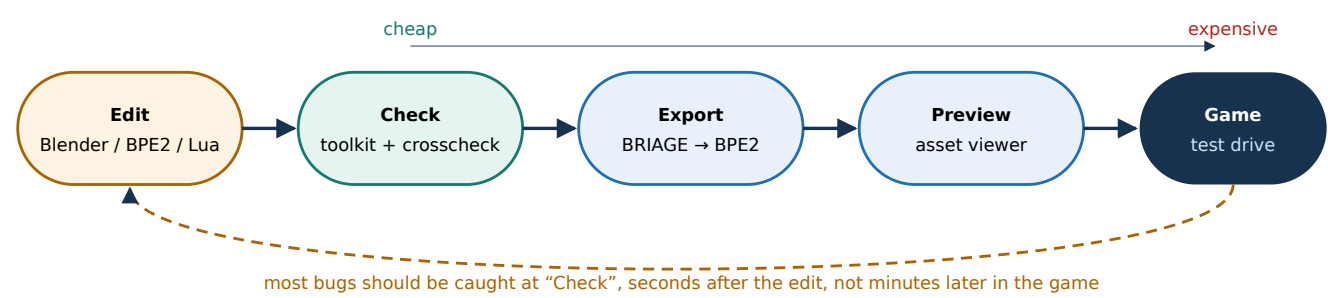


Figure 6.1 · Cost grows to the right. Move every check as far left as it will go.

You changed...	Redo
Mesh shape, UVs, materials, textures	BRIAGE IGS export → BPE2 export of the vehicle
Object names or hierarchy	IGS and every .ia (names must match) → re-check blueprint links (crosscheck) → BPE2 export
An animation	That .ia only → BPE2 export; clear the preview cache if it looks stale
Bogie/wheel/coupler position	IGS → toolkit Measure + Patch → BPE2 export
Blueprint values only	BPE2 export
Lua only	BPE2 export (compiles .out); reload the scenario
CSV curves	BPE2 export of the simulation blueprint

Close the game before exporting

Exporting into a provider/product the route editor has loaded can corrupt the route on save (BRIAGE §9.1.3). Many changes also only take effect after the vehicle is re-placed in the scenario (BRIAGE §10.6).

2 · Getting the vehicle into the game

- 1 Export from BPE2 so the compiled files exist under `Assets\Provider\Product` .
- 2 Start TSC, open a route in the editor, and in the object set filter enable your **Provider** and **Product** check (menu names vary by release).
- 3 Drag the engine (or wagon) from the rolling-stock browser onto the track. Build a short train with a matching-brake wagon.
- 4 Save as a test scenario (Free Roam or Standard) and drive it. Keep one test scenario per vehicle; re-place the vehicle after blueprint changes.
- 5 Optional: a **consist blueprint** (loco + wagons) makes the train reusable in scenarios check Quick Drive listing rules on your build.

3 • The test drive, in order

Test in this order: each step depends on the previous one, so a failure early saves you chasing symptoms later.

#	Check	If it fails, look at
1	Vehicle appears, textured, right way round	Geometry path, texture paths, front = +Y
2	Sits on the rails, not floating/sunk	Origin on rail head, wheel radius = centre height
3	Couples to the wagon without a gap	Coupling pivot/receiving matrices, coupling blueprints
4	Spawn state: brakes released, controls at sensible values	DefaultValue of each control
5	Brakes charge and release	Air system / ejectors, compressor, pressures
6	Moves on throttle; wheels spin at the right speed	Curves vs MaxForce, Powered, axle NodeIds, radius
7	Bogies turn on curves, not ahead of or behind the mesh	BogeyPivotX vs mesh
8	Every cab control clicks and moves with its value	PickTransformName, .ia range, linear keys
9	Exterior animations (pantograph, wipers, doors)	AnimSet IDs, AnimStyle, node merging
10	Gauges read correctly across their range	Control min/max vs dial calibration
11	Lights by direction and at night; numbers display	Light children, light meshes, digits set
12	LODs switch cleanly at distance	LOD distances, hidden LOD twins
13	Script features; LogMate shows no errors	Part 5
14	Reverse direction, other cab end, AI driving	Double cab, MaxPermissibleSpeed, GetIsPlayer gating

4 • Looking inside compiled files

TSC ships a converter, `serz.exe`, that turns compiled `.bin` blueprints into readable XML and back. The raw geometry dumps supplied with this project (`chcGeometry` XML with `TransformName`, `SourceLToPTTransform`, LOD controllers and materials) were produced by a conversion of this kind. Use it to:

- confirm what BPE2 actually compiled (values, paths),

- list the node names in an exported model and feed them to `tsc_crosscheck.py` (`--geo` , `--cabgeo`),
- study how stock vehicles solve a problem you are facing.

```
serz.exe "Assets\MyProvider\MyProduct\...\MyLoco.bin" → MyLoco.xml
python tools\tsc_crosscheck.py --engine MyLoco.xml --geo MyLoco.geo.xml --cabgeo
MyLoco_cab.geo.xml
```

check serz's exact command-line options on your install; it lives in the RailWorks folder.

5 • Shipping

- Ship the **Assets** files your vehicle needs (blueprints `.bin` , geometry, textures, animations, curves, `.out` scripts, sounds). Include Source only if you want others to modify it.
- Package with the TSC package manager (`.rwp`) or a zip that mirrors `RailWorks\Assets\Provider\Product` . **check**
- List dependencies (other products your blueprints reference, e.g. Kuju coupling or particle blueprints).
- Do not redistribute Kuju/DTG sample files as your own; re-author anything copied from samples.
- Ship with DEBUG alerts off and `print` spam removed.

6 • Troubleshooting: symptom → cause → fix

6.1 In Blender / BRIAGE

Symptom	Cause → fix
"No mesh or empty object found"	Nothing selected/visible for the chosen export mode → select, or unhide (BRIAGE §9.2)
"Please export to the same drive..."	Textures on another drive → move textures or use path tools (§9.3)
Wrong number of texture slots	Shader expects N slots → see the shader table in Part 2
"Inconsistent LODing", offset objects	LOD child not under LOD n−1, different base name, smaller distance, or 24-char truncation clash (§9.10)
Name changed in log	Missing/bad <code>L_DDDD_</code> prefix → rename (§10.4)
Settings seem ignored	Unapplied scale → Ctrl+A Scale, re-export with Verbose (§9.8)

6.2 In BPE2 / preview

Symptom	Cause → fix
No textures in preview	Texture path mapping mode or directory → check the IGS log (§9.4)
Missing frames of an animated texture	Only frame 1 is referenced → export the other frames manually (§6.4)
Animation doesn't play	"Animate in editor" off, merged node, name mismatch, stale cache (§9.5)

Symptom	Cause → fix
Export error on a Lua file	Syntax error → fix and re-export
File with extra dots misbehaves	BPE2 treats text after the first dot as extension → rename (§9.7)

6.3 In game

Symptom	Cause → fix
Spawns with brakes on	TrainBrakeControl default > 0 → 0
Weak or no traction	MaxForce below curve peak; unpowered bogies; throttle curve not starting at 0,0
Brakes never charge	No compressor / cut-out 0 / ejectors missing
Brake handle dead	CabControlName ≠ ControlName
Wheels sink or float	Radius ≠ modelled centre height
Bogie swings in the wrong place	BogeyPivotX ≠ mesh
Cab control not clickable	PickTransformName node missing in the cab IGS
Lamp/indicator never shows	Visibility element TransformName missing
Lever jumps or lags	Non-linear keys; keys not at first/last frame
Crash on load (newer builds)	Suspect a self-closing <LocoBrakeAssembly /> (handoff doc) and empty required lists
Script feature stops mid-session	Runtime error → LogMate
AI drives too fast / never stops	MaxPermissibleSpeed; brake performance
Crash or corruption after editing	Exported while the route editor had the product loaded (§9.1.3)